FixPart.de category structure — final result

How fixpart.de organises more than 20 million spare parts, what our three API shops do differently, and how we would rebuild the same structure in Laravel, PostgreSQL and Meilisearch.

FixPart did not build a bigger category tree. It built a product database with a curated navigation on top.

The wholesaler API supplies the facts: article numbers, brands, device types, compatibility, price and stock. FixPart adds the editorial layer — its own appliance names, 375 part categories, a five-level classification, SEO landing pages and a fit guarantee. That layer is what makes the shop feel curated; it is also the part we have to build ourselves.

1. **Two axes, not one tree.** Customers start with their appliance or with the part. Both paths use the same products.
2. **Navigation is device-first.** 105 appliance groups → 1,511 brands → model / type number, with a separate A–Z part index.
3. **URLs are a fixed matrix.** Appliance × brand × part pages are generated; breadcrumbs come from the URL, not the classification.
4. **One product, many listings.** One classification path per SKU; compatibility links create the many entry points without duplicating products.

01 The structure in numbers

AXIS	LEVEL	NODES	EXAMPLE
Device	**L1 · Appliance group**	**105**	Waschmaschine, Kühlschrank, Backofen
Device	**L2 · Brand**	**1,511**	Bosch, Siemens, Miele (site claims 2,000+)
Device	**L3 · Model / type number**	**thousands / brand**	WGG244Z40 — reached through the type-number filter, not a page level
Part	**L1 · Browsable part category**	**375**	Türgummi, Türfach, Pumpe, Kohlefilter
Part	**L2–L5 · Internal classification**	**5 levels**	Mechanisch › Gehäuseteil › Fenster › Tür

Classification sample: 33 verified nodes across 39 product pages — L1: 3, L2: 7, L3: 14, L4: 8, L5: 1. The full classification only exists in the supplier feed, so this is evidence of the model, not a complete census. FixPart markets "5,000+ categories", but only 375 have a public index page; the rest are classification leaves and generated appliance × brand × part combinations.

02 How the customer moves — and how the page is built

NAVIGATION

Appliance → brand → part → type number

The main entrance is **/gerate** with 105 appliance groups. People who know the part name use the flat A–Z index at **/ersatzteile/{part}**.

URLS & BREADCRUMBS

A fixed matrix of the two axes

/waschmaschine-ersatzteile/bosch/turgummi is the deepest example. The breadcrumb is rebuilt from the URL segments. On a product page it deliberately collapses to the leaf: **Home › Tür › product**.

FILTERS

Appliance · brand · type number · part category

Plus price, availability and condition (original / aftermarket / refurbished). The type-number picker on a brand page is the narrowest filter.

ONE PRODUCT, MANY LISTINGS

Bosch 00704286 — washing-machine door

One classification path (Mechanisch › Gehäuseteil › Fenster › Tür). Device brands Balay, Bosch, Profilo and Siemens. Appliance type Waschmaschine. **1,802 compatible models.** It appears on dozens of appliance × brand × part pages without being copied into 1,802 categories.

WHY THIS MATTERS COMMERCIALLY

One canonical product, many qualified entrances

Price and stock are updated once. There is no duplicate content. The generated landing pages carry the SEO, while the product page stays the source of truth for the customer and for Google.

03 Comparison with the three API shops

SHOP	NAVIGATION	DEPTH	OWN TAXONOMY / EDITORIAL LAYER
ersatzteilblitz.de	Gerätegruppen A–Z + Artikelgruppen A–Z; inquiries deep-link to shop.euras.com	2	none supplier dimensions only
wildbihler-shop.de	Hersteller A–Z + Geräte A–Z; brand × device treated as separate "manufacturers" (e.g. AEG-DAMPFBÜGELEISEN)	2–3	none brand and device stay raw
ersatzteile-24.com	Brand index, brand → device model lists, part pages; a repair assistant at /shop.php	3	none no renamed groups or part taxonomy
fixpart.de	105 appliance groups → 1,511 brands → 375 part categories → type number; separate A–Z part index	3 + 5	yes renamed groups, part categories, classification, SEO landings, fit guarantee

Identical in all four: part brand, device brand, device type, model / type number, article number, description and compatibility all come from the same supplier dimensions. The underlying articles are largely the same.

The difference is editorial: FixPart turns those dimensions into a shop. The other three remain databases — even when they sell the same part. The API provides the facts; the taxonomy is the differentiator.

04 What the API gives us — and what we must add

PROVIDED DIRECTLY BY THE WHOLESALER

- ✓ Article number, MPN and description
- ✓ Supplier part group / part type in supplier wording
- ✓ Device brand(s), device type(s), model / type numbers
- ✓ Compatibility list — the most valuable field
- ✓ Price and stock

WHAT WE MUST ADD (THE MOAT)

- + 105 normalised appliance groups (one name per device type)
- + 375 part categories + the five-level classification
- + Aliases and synonyms: supplier spelling → our category
- + SEO copy, landing pages, breadcrumbs and sitemaps
- + Fit guarantee and translations for DE / ES / IT

Avoid the trap of the API shops: never model "AEG-Waschmaschine" as a category. Brand and device are separate dimensions, connected to products through compatibility.

05 How we would rebuild it

1 PostgreSQL schema

categories (parent_id + materialised path), **brands**, **models**, **products**, **product_model** compatibility, **seo_landings**. The tree is small (~500 rows); compatibility can reach millions.

2 Import pipeline

Stage the raw feed (JSON + hash) → normalise via category aliases → upsert products with one primary category → attach compatibility rows → queue reindex. Nightly full import, idempotent.

3 Meilisearch

One **products** index: flat **category_ids**, **brand**, **appliance_types**, **condition**, **price**, **in_stock** as filterable facets. A separate **models** index powers "find by type number".

4 Storefront

Tree and breadcrumbs from PostgreSQL; filters and counts from Meilisearch. Generate appliance × brand × part pages, but publish only pages with at least three products — otherwise **noindex**.

Technical detail and working code: migration, importer, indexer and Meilisearch settings are in the separate code archive.

06 Indicative plan and the decisions we need

WEEKS 1–3 — Taxonomy + import
105 appliances, 375 part categories, aliases, nightly import, product & compatibility tables.

WEEKS 4–6 — Navigation + search
Appliance → brand → part pages, Meilisearch facets, type-number finder, breadcrumbs.

WEEKS 7–10 — SEO landings
Generated pages, sitemaps, internal linking, thin-content guard.

ONGOING — Expansion
DE → ES / IT, synonym tuning, zero-result monitoring, price/stock freshness.

OPEN QUESTIONS & ASSUMPTIONS

- **Wholesaler identity.** Assumed EURAS-type feed: ersatzteilblitz deep-links to shop.euras.com. Contract and field list to confirm.
- **Product-group path.** If the feed contains one, map it 1:1. If not, the 375-node list must be curated.
- **5,000 vs 375.** The gap is classification leaves and generated combinations, not 5,000 public index pages.
- **Brand coverage.** 1,511 captured vs 2,000+ claimed; some brands are probably not linked in the A–Z directory.
- **Legal.** Build from our own feed and editorial work — not from scraped FixPart content.

Deliverables: this 2-page result · presentation (PDF) · full category tree as CSV and JSON · separate code archive with Laravel migration, importer, Meilisearch indexer, settings and the captured research data. **Every number in this document is reproducible from the data files.**